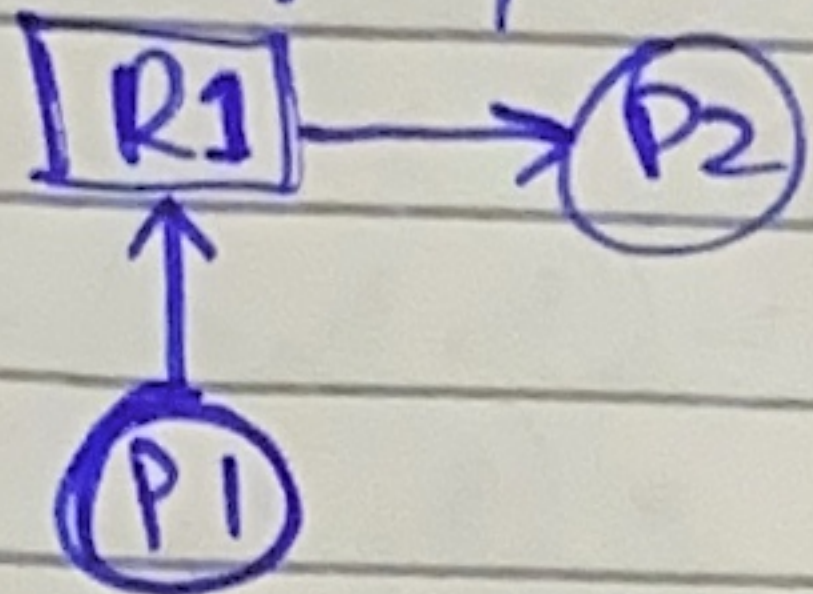


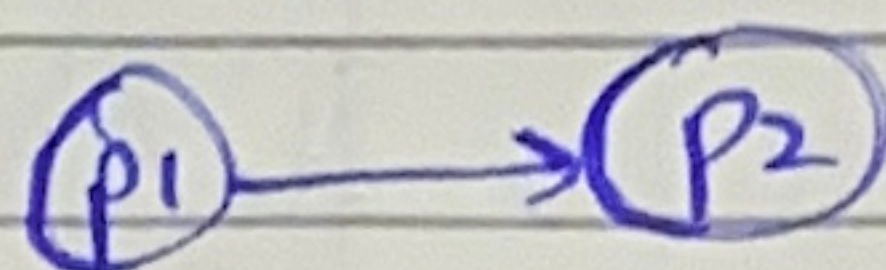
LECTURE-20

Thursday
23/4/26.

Wait - For graph: If process 1 needs Resource 1 & its waiting for R1; whereas R1 is already with Process 2; $\therefore$ P1 is indirectly waiting for P2. So, we alter the graph for minimality.



resource allocation graph



wait-for graph: P1 indirectly
waiting for P2

[3] Detection & Recovery Algo: (third method of removing Deadlocks)
from Lecture 18.

It runs after t_{n+1} to see if deadlock has occurred. It also checks if current resources could be divided among need of all processes.

- Here we don't have 'Max' matrix means we don't have info about max. resources of all processes $\therefore$ we can't calc. Ned.
- we just have info about current allocated resources ~~only~~ to all processes in Allocated matrix and Request matrix with all the resources the processes has requested already.

Working: Banker's algo was to see if the process's requested resources should be given or not considering if safe seq. exist or not for that particular req. Here we played before hand. But for Detection Algo; here the request of process is already approved & ~~has not~~ without checking safe sequence. Therefore

The detection Algo keeps on running & running and ~~also~~ during running; it is doing all the things & method of Banker's algo. Whenever the deadlock is detected; now the recovery part of algo is invoked & now we need to see which process(es) needs to be removed from system to recover deadlock.

Algo working (for detection):

same as before $\nearrow$ nxm matrix stores all the processes & resources they've requested.

	Allocation $\nearrow$ same as before			Request $\nearrow$ nxm matrix stores all the processes & resources they've requested.			Available $\rightarrow$ same as before		
	A	B	C	A	B	C	A	B	C
P ₀									
P ₁									
⋮									

* All three steps of Banker's Algo will be used on these three ~~matrix~~ in the same way. The only difference is that Need matrix is replaced with Request matrix. The new state will also be created similarly with 3 formulas & safety Algo.

$\Rightarrow$ Recovery Algo: (After Detection)

- Abort all deadlocked processes. This will be done by aborting processes one by one until deadlock removed.

The order of aborting will be decided by:

- 1) Priority of process
- 2) How long process has computed; & how much more req.
- 3) Resources the process has used.
- 4) Resources process needs to be completed.
- 5) How many processes will be terminated.

- Select a victim - minimises cost

- Roll Back - Return to some safe state, restart process for that state

- Starvation - same process may always be picked as a victim, include no. of rollbacks in cost factor.

CHAPTER: Memory Management

⇒ Logical Address space: Sort of a set in which there are all the logical addresses of all the things related to suppose Process 1; that'll be logical address space of P1. All the addresses in it has offsets for mapping purpose.

⇒ Physical Address Space: ^{Address of} ↑ is jagah par RAM main actually Process 1 (and uski instruction) pari ho

⇒ Physical & logical Address: One single line (which has suppose one instruction of P1) in logical address space has logical address. And logical address getting mapped & telling where actually that specific instruction is in the physical address space tells that it is its physical address.